
Compositional Separation Predicts What Protein Language Model Steering Can and Cannot Do

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Abstract

1 Activation steering edits a protein language model’s residual stream at inference
2 time and can change its generated sequences without retraining, but a higher proxy
3 score does not by itself establish control of a biological property, and the field
4 lacks an *a priori* rule for which properties a given steering method can move at all.
5 We attempted to steer nine sequence properties in ESM2-650M under one fixed
6 harness—mean-pooled per-layer difference-of-means vectors added uniformly at
7 every position, evaluated by single-shot masked filling—and one pre-registered six-
8 criterion acceptance protocol. A single quantity measured before any generation,
9 the mean amino-acid *compositional separation* between the low- and high-label
10 groups used to build the steering vector, ordered every outcome: it predicts both
11 whether a target is steerable and how large the achievable effect is, whereas the
12 proxy’s correlation with the true label does not. A binding proxy validated at held-
13 out $r = 0.80$ steered nothing (separation 0.003), while a catalytic proxy at $r = 0.22$
14 (separation 0.02–0.05) passed on all three seeds; effect size grew with separation
15 from a flat null to 72% of the natural property gap. Separation is necessary but not
16 sufficient: two well-separated targets still failed, one because the model does not act
17 on the direction and one because the property is *positional*. That last case exposes
18 a structural limit: uniform-addition steering can move aggregate compositional
19 properties but cannot install positionally-localized features, regardless of proxy
20 quality. All endpoints are computational; no wet-lab assay was run.

21 1 Problem

22 Protein language models learn statistical structure from large sequence collections and support
23 prediction and generation [Lin et al., 2023]. Activation steering modifies their internal representations
24 at inference time: a direction is estimated from the difference between activations for high-label and
25 low-label examples, then added to the residual stream during generation. Prior work applied this
26 method to protein generation and optimization [Huang et al., 2025].

27 Two problems sit between a steering run and a useful conclusion. First, a generated sequence can
28 receive a higher proxy score for reasons unrelated to the claimed property—by repeatedly inserting
29 residues that the scoring rule rewards, or after low-complexity outputs have been silently dropped
30 from the comparison. This half of the problem is an evaluation-validity question, and prior negative-
31 results work has catalogued how each such check can overturn an apparent success. Second, and less
32 studied, is a *selection* question: given a fixed steering method, which target properties can it move
33 at all, and can that be predicted before spending any compute? A practitioner who can only learn
34 steerability by running the full harness on each candidate has no way to budget effort or to know
35 when a null is a property of the method rather than of the target.

This paper reports what a single arc of nine steering attempts on ESM2-650M revealed about the second question. The central finding is a predictor and a structural limit, and both are, in the venue’s sense, failure results: most well-separated candidates still fail, and the method is provably unable to reach an entire class of biologically important properties.

2 Proposed Approach

2.1 One fixed harness

Every target uses the same intervention and evaluation, so that outcomes are comparable across properties. The model is ESM2-650M. Generation uses single-shot masked filling (`mask_fill_generate`, mask fraction 0.3, argmax decode). The steering direction is the per-layer difference between mean activations of 150 high-label and 150 low-label proteins, added to all 33 layers uniformly across sequence positions. The control is a matched-norm random direction. Each run evaluates 150 held-out proteins; a target-specific compositional proxy scores each generation, and the reported effect is the paired-bootstrap contrast (10,000 resamples) between the real and random directions.

The masked-fill evaluation collapses into degenerate low-complexity output at steering strength $\alpha \geq 0.35$; dose-response is therefore measured on the non-degenerate grid $\{0.10, 0.15, 0.20, 0.25\}$. This caps the achievable effect but keeps every comparison inside a valid operating window.

2.2 A pre-registered acceptance protocol

To separate real capability from a flattering scalar, each target was judged against six criteria fixed in advance. A capability gain counts only if it (1) beats the random-direction control with a real paired-bootstrap confidence interval; (2) shows a coherent dose-response over at least three strengths; (3) survives *residue exclusion*—the effect must remain significant after the two dominant substituted residues are removed from the score; (4) uses a proxy validated against real experimental labels *before* the run, asserted in code at $|r| \geq 0.15$; (5) beats any prior technique for the property; and (6) is adequately powered ($n \geq 150$). Failing (1), (2), (3), or (4) outright is a KILL; passing most but failing a dose check or (3) is AMBIGUOUS. Criterion (3) is deliberately strict because every proxy here is a function of amino-acid composition, so steering can raise it simply by enriching favored residues.

2.3 A compositional-separation gate, promoted to a predictor

Before any generation, we measure the mean per-residue amino-acid-composition L_2 distance between the low- and high-label groups from which the vector is built. This gate was introduced to explain one early null and became the arc’s most predictive number. The claim tested across the nine targets is that this pre-run separation, not the proxy’s label correlation, orders both steerability and effect size.

2.4 Targets and data

Nine properties were attempted, chosen to vary compositional separation and to contrast intrinsic against relational properties. Transmembrane fraction, DNA-binding propensity, catalytic turnover, glycosylation density, disulfide content, signal peptide, zinc-finger and calcium-binding domains were drawn from reviewed UniProt [The UniProt Consortium, 2023], cross-protein k_{cat} from DLKcat [Li et al., 2022], single-backbone binding from ProteinGym [Notin et al., 2023], and disorder from DisProt [Quaglia et al., 2022]. Proxies use published biophysical scales: mean Kyte–Doolittle hydropathy for membrane character [Kyte and Doolittle, 1982], Lys+Arg fraction for nucleic-acid interfaces [Luscombe et al., 2001], a glycine–arginine contrast for the activity–stability tradeoff in turnover [Siddiqui and Cavicchioli, 2006], and the TOP-IDP scale for disorder [Campen et al., 2008]. An earlier immune-endpoint target using IEDB [Vita et al., 2019] and a solubility target using eSol [Niwa et al., 2009] were stopped by the validity checks and are discussed only where they inform the selection question.

3 Observed Outcome

3.1 Separation orders steerability and effect size

Table 1 lists all nine targets sorted by verdict. Effect size tracks pre-run compositional separation monotonically, while the proxy’s label correlation does not. The two anchoring rows are the sharpest: the single-backbone binding target has the strongest-validated proxy in the entire arc (held-out $r = 0.80$) and produces a clean null at every strength, because its low- and high-label groups are one- and two-residue point mutants of one backbone and are therefore nearly compositionally identical (separation 0.003). Catalytic turnover has the weakest runnable proxy ($r = 0.22$) yet passes on all three seeds. Proxy-validation strength and steering effect are not even weakly monotonically related.

Table 1: Nine steering targets on ESM2-650M under one fixed harness, sorted by verdict. Separation is the pre-run compositional L_2 distance between vector groups; proxy r is the held-out label correlation; effect is expressed, where a natural low→high proxy gap is defined, as the fraction of that gap reached at $\alpha = 0.25$. Effect size follows separation, not proxy r .

Target	Sep.	Proxy r	Verdict	Effect / failure mode
Binding, single-backbone	0.003	0.80	KILL	flat null (no separation)
Zinc-finger domain	0.028	0.29	KILL	direction anti-correlated
Calcium-binding (EF-hand)	0.038	0.16	KILL	model inert to direction
Signal peptide	0.035	0.74	KILL	positional, not compositional
Disulfide (Cys)	0.026	0.17	pass (2/3)	tiny (~ 0.002)
Glycosylation density	0.039	0.41	pass (3/3)	tiny (~ 0.003)
DNA-binding propensity	0.039	0.25	pass (2/3)	25% of natural gap
Catalytic turnover (k_{cat})	0.02–0.05	0.22	pass (3/3)	moderate
Transmembrane fraction	0.086	0.79	pass (3/3)	72% of natural gap

The calibration is visible along the passing rows: binding at separation 0.003 gives a null; DNA-binding at 0.039 moves generations $\sim 25\%$ of the way to the binder distribution; transmembrane at 0.086 moves them $\sim 72\%$ of the way to the membrane-protein distribution ($\alpha = 0.25$ proxy $-0.363 \rightarrow +0.266$ against a natural gap of 0.871). Moving the binding target from single-backbone to cross-protein data (DNA-binding) lifts separation $\sim 12\times$ and converts the null into a significant, residue-robust, dose-responsive effect. The earlier single-backbone null was thus an artifact of the data geometry, not a property of binding, and the recovery was predicted from separation before any GPU run.

3.2 Separation is necessary but not sufficient

Screening by separation removes the null mode but does not guarantee success. Two targets with separation as high as the passing band still failed, by distinct mechanisms. Calcium-binding (separation 0.038) shows no significant steering at any safe strength: a compositionally distinct group whose direction the residual stream simply does not act on. Zinc-finger’s cysteine–histidine direction becomes *anti-correlated* with its proxy after steering. High separation is a prerequisite, but the model must also respond to the direction, and the direction must encode the property rather than a confound.

3.3 The harness steers composition, not position

The signal-peptide target is the sharpest failure. Its proxy—mean hydropathy over the N-terminal 30-residue window—is the best-validated of any candidate in the arc (held-out $r = 0.742$), and it produces a textbook dose-response (N-terminal hydropathy $0.059 \rightarrow 0.100 \rightarrow 0.152 \rightarrow 0.251$ across the safe grid). It nonetheless KILLS on criterion (3): excluding the two dominant substituted residues collapses the effect to non-significance. The steering vector raises N-terminal-window hydropathy purely by enriching bulk hydrophobic residues across the *whole* sequence; it cannot *concentrate* them at the N-terminus, which is what defines a signal peptide. Glycosylation density, by contrast, passes precisely because its proxy is a sequon count summed over the whole sequence—a genuinely aggregate quantity a mean-pooled vector can nudge.

115 4 Reason for Failure

116 4.1 Why proxy correlation misleads and separation predicts

117 A difference-of-means direction can encode only what differs, on average, between the two groups it
118 is built from. When the groups are near-identical in composition—single-backbone point mutants—
119 the direction carries almost no signal, no matter how faithfully the *scoring* function tracks the real
120 label. Proxy correlation certifies the ruler; separation certifies that there is something for the vector to
121 push. This is why the best-validated proxy in the arc produced the cleanest null, and why a weak
122 proxy over compositionally separated groups produced a replicated pass. Effect size then scales with
123 separation because a larger mean gap gives the added direction more magnitude to move generations
124 along.

125 4.2 Why well-separated targets still fail

126 Separation is a property of the data; steerability additionally requires that the model’s residual stream
127 respond to the direction and that the direction capture the property rather than a correlate. Calcium-
128 binding fails the first: the acidic-residue direction is well-separated but inert under this intervention.
129 Zinc-finger fails the second: its direction moves against its own proxy. Neither failure is visible from
130 separation or proxy correlation alone, so the pre-registered head-to-head control (criterion 1) and
131 residue-exclusion (criterion 3) remain necessary even after the separation gate is cleared.

132 4.3 Why an entire property class is unreachable

133 The signal-peptide result is not a tuning failure but a structural one. A vector that is mean-pooled
134 over positions and added identically to every position can change the *amount* of a residue type in
135 a sequence but not *where* it goes. Aggregate compositional properties—transmembrane fraction,
136 catalytic composition, DNA-binding charge, glycosylation density—are functions of that amount and
137 are reachable. Positionally-localized features—a signal peptide, and by the same argument active-site
138 geometry, domain boundaries, and termini-specific motifs—are functions of arrangement and are not.
139 Reaching them would require position-aware injection, a different intervention, not a different target.
140 This bounds the class of biological properties this widely used steering recipe can address.

141 4.4 The validity checks that make the predictor trustworthy

142 The predictor is only credible because the acceptance protocol rejected flattering numbers elsewhere
143 in the arc, and each rejection changed a conclusion. An immune-endpoint proxy correlated $r = 0.427$
144 with an allele-aggregated binding label but only 0.100 with measured T-cell response, and its full-
145 length correlation fell from +0.379 under random folds to -0.323 under organism-grouped folds—so
146 it was stopped before generation for measuring the wrong endpoint. A layer-subset thermostability
147 comparison looked competitive only because the all-layer arm’s outputs had collapsed under the
148 low-complexity filter, leaving too few evaluable pairs; inside the shared valid range the subset retained
149 just 43–59% of the full effect. A solubility contrast of +0.0125 fell to +0.00035, with a confidence
150 interval crossing zero, once its two dominant substituted residues were excluded. A disorder effect
151 repeated across three stored runs but survived residue exclusion in only two. Each check protects a
152 different inference: endpoint validity, evaluable denominator, composition dependence, and whole-
153 run stability. Only after a target clears all of them does its separation-versus-effect placement in
154 Table 1 carry weight.

155 5 Discussion and Limitations

156 The useful consequence is that, for this harness, steerability and its ceiling are predictable before
157 any compute from the compositional separation of the vector-building groups, and an entire property
158 class (positional features) is ruled out on structural grounds. The productive next steps are therefore
159 not to try more properties but to change the intervention—position-aware injection—and to validate
160 the strongest compositional gain against a biological assay.

161 The evidence has firm limits. Every endpoint is computational: no wet-lab or structural assay
162 establishes that any generated sequence has improved thermostability, turnover, transmembrane

character, binding, or glycosylation; each “effect” is a change in a compositional proxy. All results use one model (ESM2-650M), one decoder (single-shot masked filling), and one injection geometry (uniform difference-of-means addition), so the composition-only limit is specific to that geometry. The usable dose range is narrow because the evaluation degenerates at $\alpha \geq 0.35$, so passes are demonstrated at modest strength and achievable effect is capped by the evaluation rather than necessarily by the direction. Proxies are compositional and non-mechanistic; a pass shows the model can be pushed toward an amino-acid signature that correlates with the property, not that mechanistic specificity is preserved. Several validity checks were clarified after outcomes were seen, bootstrap intervals describe stability over the fixed evaluation proteins rather than population guarantees, and some legacy runs omit seed metadata. The separation predictor is calibrated on nine targets from one arc and one model; whether the same ordering holds for other models, decoders, or injection geometries is untested.

LLM usage disclosure

Large language models assisted with manuscript restructuring, prose editing, and source checks, and with code and earlier drafting during the underlying study. The authors made the scientific decisions and verified every numerical claim against the saved artifacts.

Code and data availability

The study uses public protein resources (UniProt, DLKcat, ProteinGym, DisProt, IEDB, eSol) together with saved model outputs. An anonymized artifact package with per-claim source files and seed details will be provided if the submission system supports it. The review manuscript does not link to an identifying repository.

Ethics declaration

This computational study used public protein sequences and aggregated public annotations. It involved no new human participants, clinical intervention, or collection of personal data. Generated sequences were used only to compute compositional proxy scores and were not synthesized or expressed.

Reproducibility statement

Each numerical claim maps to a saved result artifact and a study record fixing the dataset, proxy, separation, seeds, and dose grid. The intervention, controls, and acceptance criteria were fixed before the runs they judge.

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